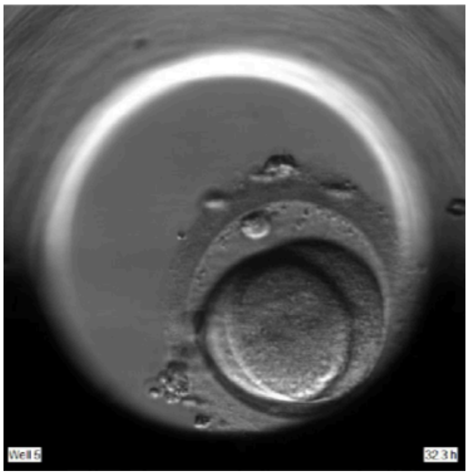
Use Case 2: Segmentation of lung cancer cells in microscopy images



You are part of a research team developing an AI model for automatic segmentation of lung cancer cells in microscopy images. The goal is to support quantitative analysis of cellular morphology and assist researchers in studying tumor characteristics at the single-cell level.

The model predicts a segmentation mask for each image, where every pixel is assigned either to a cancer cell or to the surrounding background. Each target structure is expected to correspond to a single, compact cell without overlapping labels or disconnected components.

Researchers are particularly interested in obtaining accurate delineation of cell boundaries, since small boundary errors may substantially affect downstream morphological analysis. Precise estimation of cell volume, however, is considered less important for the intended application.

The reference annotations were created by multiple experts. During annotation review, the team notices that different annotators occasionally disagree on the exact cell borders, especially in regions with weak contrast or blurred boundaries.

Some algorithms under evaluation occasionally fail on difficult images and may produce empty segmentation masks or corrupted outputs. In addition, isolated prediction artifacts away from the actual cells are considered problematic, even when they are small.

Your team plans to report results in a biomedical imaging publication and wants to use metrics that are commonly established and easily interpretable within the medical imaging community.

You now want to validate the segmentation model appropriately and decide which metrics should be used for reporting model performance.

Task description

Study your use case carefully. Then go to: https://metrics-reloaded.dkfz.de/metric-selection

Follow the decision process of the Metrics Reloaded toolkit and answer the fingerprint questions based on the information provided in the given scenario. Your goal is to identify the most appropriate performance metrics for the given application.

After finishing the task, save the generated PDF report containing the selected metrics and fingerprint answers. Be prepared to briefly present your use case to the group, including:

- how you interpreted the scenario,
- which fingerprint questions were easy or difficult to answer,
- whether any design choices were unclear or ambiguous,
- and which metrics were ultimately selected by the toolkit.

Please send your PDF report to the lecturers after completing the exercise.